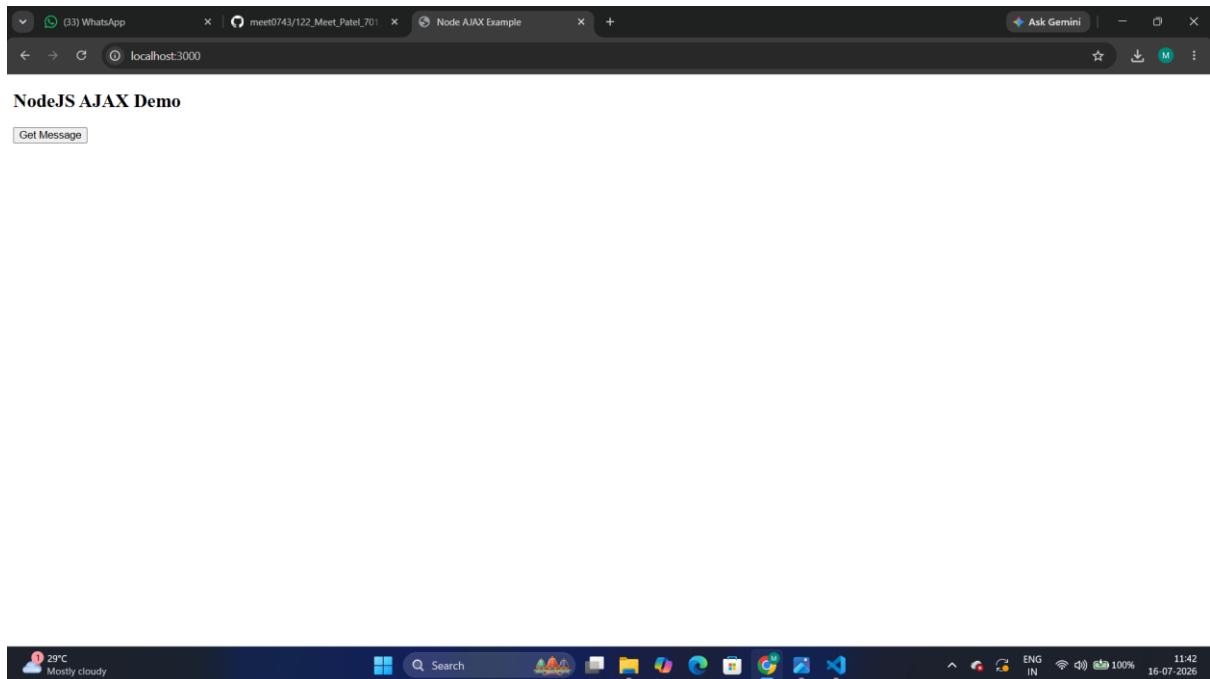
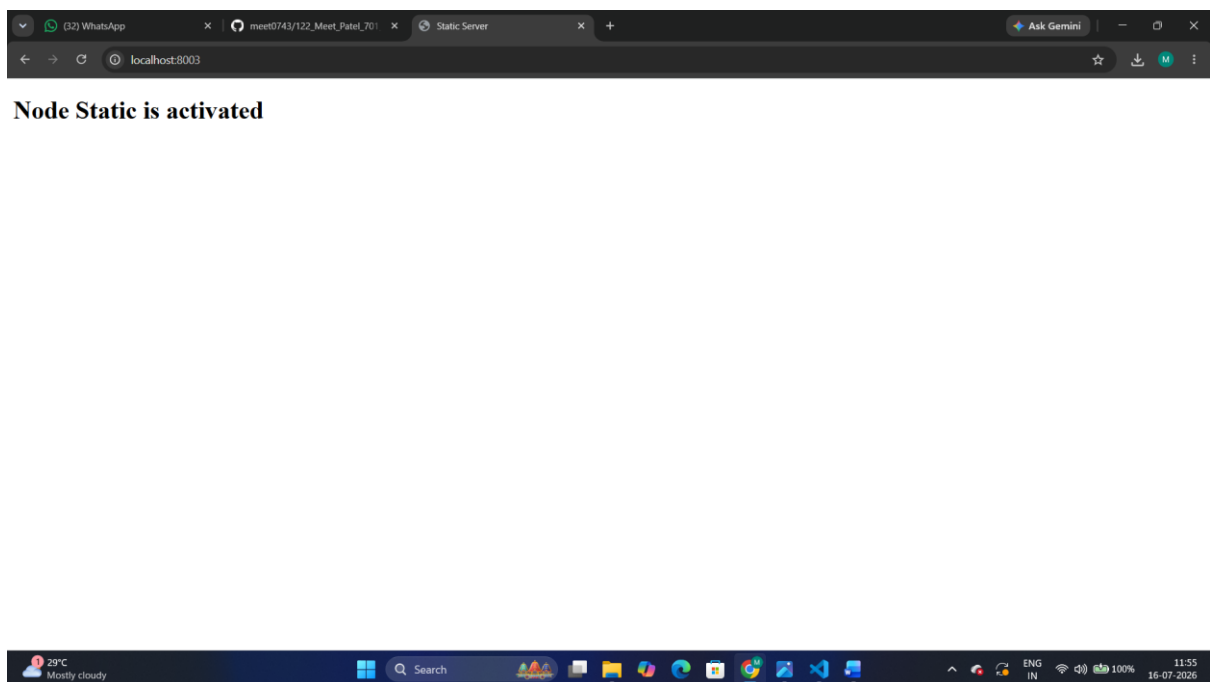


Q1:



Q2:



Q3:

The screenshot shows a VS Code editor with a project named 'assignment_1'. The Explorer sidebar on the left lists files: 'ASSIGNMENT_1', 'que1', 'que2', 'que3', 'app.js', 'chatbot.js', 'package.json', 'que4', 'que5', 'que6', 'que7', 'que8', 'que9', 'que10', 'que11', 'que12', and '.gitignore'. The 'app.js' file is open in the editor, showing a JavaScript function 'reply' that handles chatbot responses based on keywords like 'you', 'where', 'live', and 'fine'. The terminal at the bottom shows the command 'node app.js' being executed, and the chatbot's responses are displayed: 'Hi!', 'I'm 25', 'How are you', 'I'm fine ^_^', 'I live in India', and 'Sorry, I didn't get it :('.

```
que3 > JS app.js > reply
17 function reply(message)
18 {
19   message.indexOf("you")
20 }
21
22
23
24
25
26
27
28
29
30
31
32
33
34
35
36
37
38
39   return "I'm fine ^_^"
40 }
41
42
43   else if(message.indexOf("where") > -1
44     && message.indexOf("live") &&
45     message.indexOf("you"))
46   {
47     return "I live in " + this.Bot_Country;
48   }
49   return "Sorry, I didn't get it :( ";
50 }
51 }
```

```
PS D:\full_stack\assignment_1> cd que3
PS D:\full_stack\assignment_1\que3> node app.js
You==>Hi
Bot ==> Hi!
You==>age
Bot ==> I'm 25
You==>how are you
Bot ==> I'm fine ^_^
You==>where live you
Bot ==> I live in India
You==>
Bot ==> Sorry, I didn't get it :(
You==>
```

Q4:

The screenshot shows a VS Code editor with a project named 'assignment_1'. The Explorer sidebar on the left lists files: 'ASSIGNMENT_1', 'que1', 'que2', 'que3', 'files', 'node_modules', 'package-lock.json', 'package.json', 'zipfolder.js', 'que5', 'que6', 'que7', 'que8', 'que9', 'que10', 'que11', 'que12', and '.gitignore'. The 'zipfolder.js' file is open in the editor, showing a JavaScript function 'reply' that handles chatbot responses based on keywords like 'you', 'where', 'live', and 'fine'. The terminal at the bottom shows the command 'node zipfolder.js' being executed, and the chatbot's responses are displayed: 'Hi!', 'I'm 25', 'How are you', 'I'm fine ^_^', 'I live in India', and 'Sorry, I didn't get it :('.

```
que3 > JS app.js > reply
17 function reply(message)
18 {
19   message.indexOf("you")
20 }
21
22
23
24
25
26
27
28
29
30
31
32
33
34
35
36
37
38
39   return "I'm fine ^_^"
40 }
41
42
43   else if(message.indexOf("where") > -1
44     && message.indexOf("live") &&
45     message.indexOf("you"))
46   {
47     return "I live in " + this.Bot_Country;
48   }
49   return "Sorry, I didn't get it :( ";
50 }
51 }
```

```
PS D:\full_stack\assignment_1> cd que4
PS D:\full_stack\assignment_1\que4> node zipfolder.js
File Compressed.
PS D:\full_stack\assignment_1\que4>
```

Q5:

```
17 function reply(message)
18 {
19   message.indexOf("you")
20 }
21
22 {
23   return "I'm fine ^_^"
24 }
25
26 else if(message.indexOf("where") > -1
27   && message.indexOf("live") &&
28   message.indexOf("you"))
29 {
30   return "I live in " + this.Bot_Country;
31 }
32
33 return "Sorry, I didn't get it :( ";
34 }
35 }
```

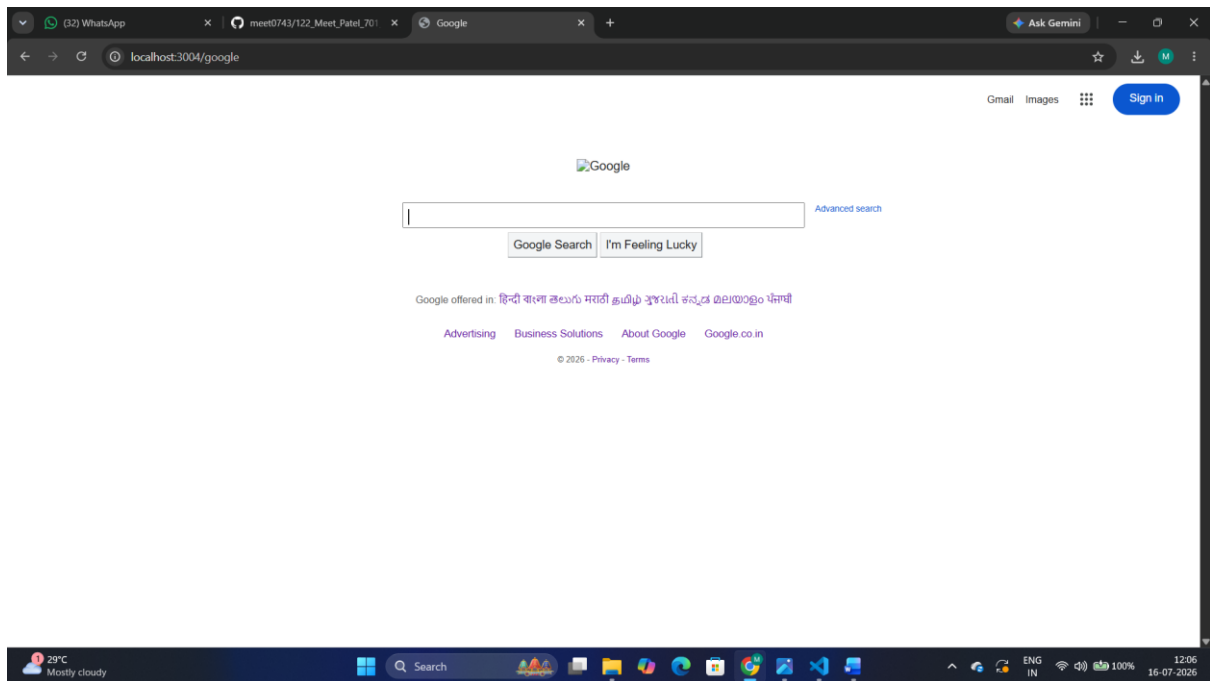
```
PS D:\full_stack\assignment_1> cd que5
PS D:\full_stack\assignment_1\que5> node extractZip.js
File Decompressed.
PS D:\full_stack\assignment_1\que5>
```

Q6:

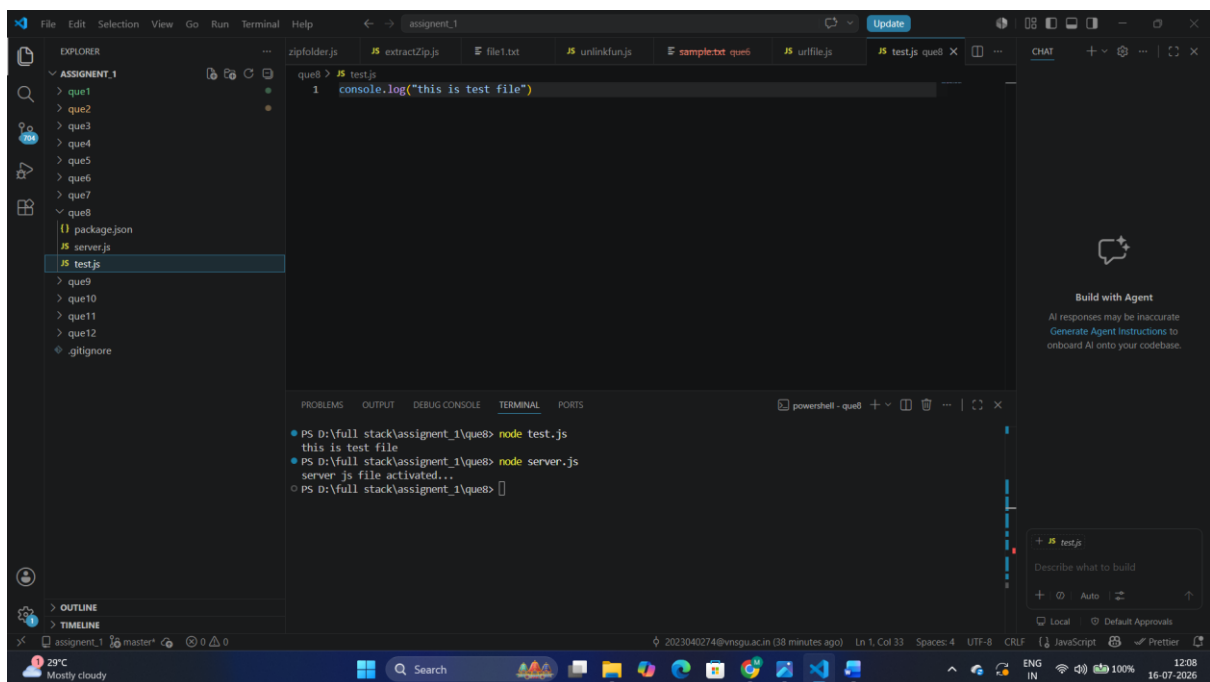
```
1 const fs = require("fs");
2 const util = require("util");
3
4 // Promisify fs.unlink
5 const unlinkFile = util.promisify(fs.unlink);
6
7 // Delete file
8 unlinkFile("sample.txt")
9 .then(() => {
10   console.log("File deleted successfully.");
11 })
12 .catch((err) => {
13   console.log("Error:", err.message);
14 });
```

```
PS D:\full_stack\assignment_1\que6> node unlinkfun.js
File deleted successfully.
PS D:\full_stack\assignment_1\que6>
```

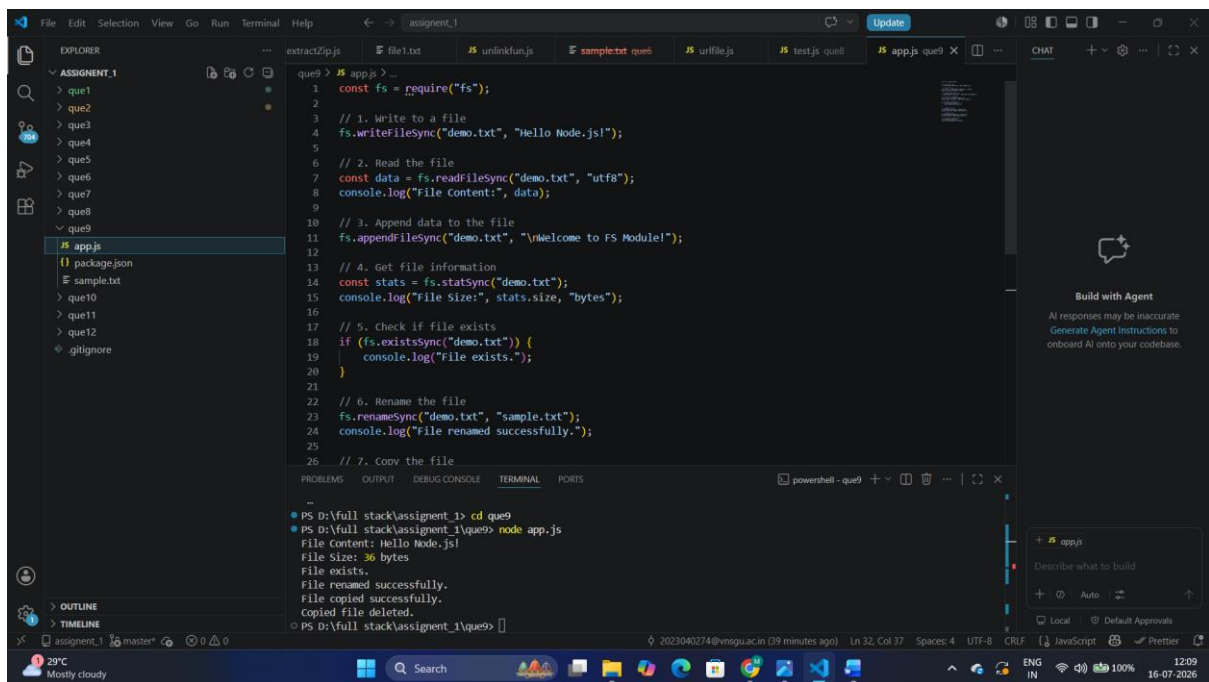
Q7:



Q8:



Q9:



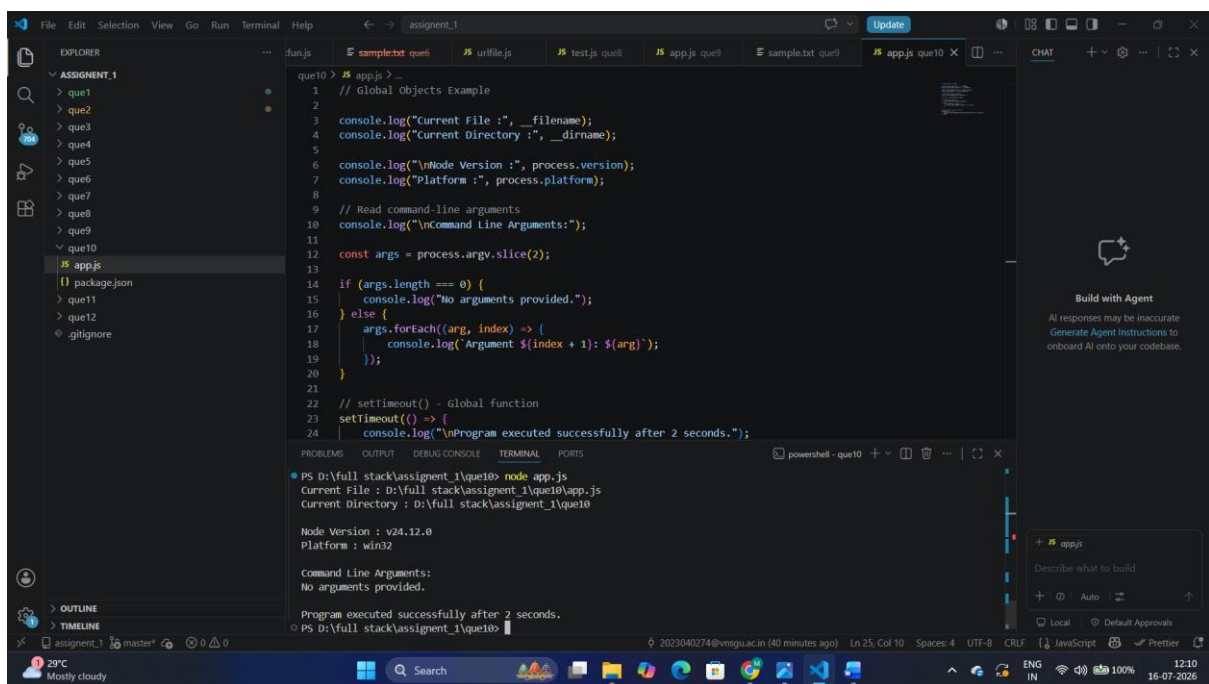
The screenshot shows a VS Code editor window with a file explorer on the left displaying a project named 'ASSIGNMENT_1' containing files 'package.json', 'sample.txt', and a directory 'que9'. The main editor displays 'app.js' with the following code:

```
1 const fs = require("fs");
2
3 // 1. Write to a file
4 fs.writeFileSync("demo.txt", "Hello Node.js!");
5
6 // 2. Read the file
7 const data = fs.readFileSync("demo.txt", "utf8");
8 console.log("File Content:", data);
9
10 // 3. Append data to the file
11 fs.appendFileSync("demo.txt", "\nWelcome to FS Module!");
12
13 // 4. Get file information
14 const stats = fs.statSync("demo.txt");
15 console.log("File Size:", stats.size, "bytes");
16
17 // 5. Check if file exists
18 if (fs.existsSync("demo.txt")) {
19     console.log("File exists.");
20 }
21
22 // 6. Rename the file
23 fs.renameSync("demo.txt", "sample.txt");
24 console.log("File renamed successfully.");
25
26 // 7. Copy the file
```

The terminal window at the bottom shows the execution of 'node app.js' in the directory 'D:\full_stack\assignment_1\que9'. The output is:

```
PS D:\full_stack\assignment_1\que9> node app.js
File Content: Hello Node.js!
File Size: 36 bytes
File exists.
File renamed successfully.
File copied successfully.
Copied file deleted.
```

Q10:



The screenshot shows a VS Code editor window with a file explorer on the left displaying a project named 'ASSIGNMENT_1' containing files 'package.json', 'sample.txt', and a directory 'que10'. The main editor displays 'app.js' with the following code:

```
1 // Global Objects Example
2
3 console.log("Current File :", __filename);
4 console.log("Current Directory :", __dirname);
5
6 console.log("\nNode Version :", process.version);
7 console.log("Platform :", process.platform);
8
9 // Read command-line arguments
10 console.log("\nCommand Line Arguments:");
11
12 const args = process.argv.slice(2);
13
14 if (args.length === 0) {
15     console.log("No arguments provided.");
16 } else {
17     args.forEach((arg, index) => {
18         console.log(`Argument ${index + 1}: ${arg}`);
19     });
20 }
21
22 // setTimeout() - Global function
23 setTimeout(() => {
24     console.log("\nProgram executed successfully after 2 seconds.");
25 }, 2000);
```

The terminal window at the bottom shows the execution of 'node app.js' in the directory 'D:\full_stack\assignment_1\que10'. The output is:

```
PS D:\full_stack\assignment_1\que10> node app.js
Current File : D:\full_stack\assignment_1\que10\app.js
Current Directory : D:\full_stack\assignment_1\que10

Node Version : v24.12.0
Platform : win32

Command Line Arguments:
No arguments provided.

Program executed successfully after 2 seconds.
```

Q11:

The screenshot shows the VS Code editor with a project named 'assignment_1'. The Explorer sidebar on the left shows a file structure with 'que1' through 'que11', 'index.js', 'package.json', 'test.js', 'que12', and '.gitignore'. The main editor window displays the code for 'test.js' in the 'que11' tab. The code is as follows:

```
1 const greet = require("./index");
2
3 console.log(greet("Meet"));
```

The terminal at the bottom shows the command 'node test.js' being executed, resulting in the output 'Hello, Meet! Welcome to my npm package.'.

On the right side, there is a 'Build with Agent' panel with a message: 'All responses may be inaccurate. Generate Agent Instructions to onboard AI onto your codebase.'

Q12:

The screenshot shows the VS Code editor with a project named 'assignment_1'. The Explorer sidebar on the left shows a file structure with 'que1' through 'que12', 'index.html', 'public', 'app.js', 'server.js', 'inc', 'package.json', and '.gitignore'. The main editor window displays the code for 'app.js' in the 'que12' tab. The code is as follows:

```
1 const fs = require("fs").promises;
2
3 // Task 1: Create a file
4 fs.writeFile("demo.txt", "Hello Node.js!")
5   .then(() => {
6     console.log("Task 1: File created.");
7   })
8   // Task 2: Read the file
9   .then(() => {
10    return fs.readFile("demo.txt", "utf8");
11  })
12  .then(data => {
13    console.log("Task 2: File content:", data);
14  })
15  // Task 3: Delete the file
16  .then(() => {
17    return fs.unlink("demo.txt");
18  })
19  .then(() => {
20    console.log("Task 3: File deleted.");
21  });
```

The terminal at the bottom shows the command 'node app.js' being executed, resulting in the output 'Task 1: File created.', 'Task 2: File content: Hello Node.js!', and 'Task 3: File deleted.'

On the right side, there is a 'Build with Agent' panel with a message: 'All responses may be inaccurate. Generate Agent Instructions to onboard AI onto your codebase.'